

# Harry Reyes Nieva

Columbia University Irving Medical Center  
Division of Infectious Diseases  
622 West 168th Street, PH9-901  
New York, NY 10032

Email: [harry.reyes@columbia.edu](mailto:harry.reyes@columbia.edu)  
Homepage: <https://www.harryreyesnieva.com>

## Education

- Columbia University in the City of New York, New York, NY  
*Doctor of Philosophy* in Biomedical Informatics, conferred February 2025  
*Master of Philosophy* in Biomedical Informatics, 2023  
*Master of Arts* in Biomedical Informatics, 2021 (GPA: 4.0)  
Dissertation Committee: Noémie Elhadad (doctoral advisor), Suzanne Bakken, Karthik Natarajan, Jason Zucker, Li Zhou (Harvard Medical School)  
Coursework included: Machine Learning for Healthcare, Natural Language Processing, Data Science, Symbolic AI for Healthcare, Computer Applications in Healthcare, Intelligent Decision Systems
- Beijing Language and Culture University, Beijing, China  
*Visiting Student*, Intensive Mandarin, 2019
- Johns Hopkins Bloomberg School of Public Health, Baltimore, MD  
*Master of Applied Science* in Spatial Analysis for Public Health, 2019  
Coursework included: Biostatistics, Spatial Statistics, Epidemiology, Spatial Data Technologies
- Harvard T.H. Chan School of Public Health, Boston, MA  
*Special Student*, International and Population Health, 2009-2011  
Coursework included: Immunology, Tuberculosis, Reproductive Health Issues, Survey Research Methods, Planning and Evaluation of Health Programs, Financing Health Care in Developing Countries
- Yale University, New Haven, CT  
*Bachelor of Arts* in Sociology and History, 2005  
Coursework included: Elements of Computing, Current Issues in Biological Sciences, General Chemistry, Social Policy Analysis, Portuguese for Spanish Speakers, Elementary Yoruba

## Faculty Academic Appointments

|                 |                                    |                        |                                                         |
|-----------------|------------------------------------|------------------------|---------------------------------------------------------|
| 06/2026-present | Lecturer in Biomedical Informatics | Biomedical Informatics | Columbia University Irving Medical Center, New York, NY |
|-----------------|------------------------------------|------------------------|---------------------------------------------------------|

## Postgraduate Training

|                 |                                 |                              |                                                          |
|-----------------|---------------------------------|------------------------------|----------------------------------------------------------|
| 06/2026-present | Postdoctoral Research Scientist | Artificial Intelligence      | Columbia Core AI Lab, New York, NY                       |
| 02/2026-present | Trainee Associate Member        | Medicine Oncology            | Herbert Irving Comprehensive Cancer Center, New York, NY |
| 04/2025-present | Postdoctoral Research Scientist | Medicine Infectious Diseases | Columbia University Irving Medical Center, New York, NY  |

|                 |                                                                                                                                 |                                   |                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------|
| 09/2022-08/2024 | National Library of Medicine/<br>National Institute of Allergy and Infectious Diseases T15 Predoctoral Research Training Fellow | Biomedical Informatics            | Columbia University Irving Medical Center, New York, NY |
| 09/2019-08/2022 | National Library of Medicine T15 Predoctoral Research Training Fellow                                                           | Biomedical Informatics            | Columbia University Irving Medical Center, New York, NY |
| 08/2013-10/2023 | Visiting Postgraduate Research Fellow                                                                                           | Medicine                          | Harvard Medical School, Boston, MA                      |
| 08/2013-08/2019 | Predoctoral Research Fellow                                                                                                     | General Medicine and Primary Care | Brigham and Women's Hospital (BWH), Boston, MA          |

## Other Professional Positions

|           |                                           |                                                                                 |
|-----------|-------------------------------------------|---------------------------------------------------------------------------------|
| 2025      | Staff Associate                           | Department of Biomedical Informatics, Columbia University Irving Medical Center |
| 2012-2019 | Research Manager                          | Division of General Internal Medicine and Primary Care, BWH                     |
| 2010-2012 | Data Quality Assurance Manager            | Harvard University/U.S. President's Emergency Plan for AIDS Relief (PEPFAR)     |
| 2008-2010 | Data, Training, and Education Coordinator | Harvard University/U.S. President's Emergency Plan for AIDS Relief (PEPFAR)     |
| 2007-2008 | Research Assistant                        | Division of General Internal Medicine and Primary Care, BWH                     |
| 2006-2007 | Emergency Medical Technician              | Tufts Emergency Medical Services                                                |
| 2005-2006 | Teacher, English-as-a-Second-Language     | New York City Board of Education/New York City Teaching Fellows                 |
| 2004-2005 | Client Outreach Officer                   | Advanced Strategies for Healthcare Access                                       |
| 2004      | Research Assistant                        | Department of Sociology, Yale University                                        |
| 2003-2004 | Editorial Assistant/Contributor           | Yale Medicine Magazine                                                          |
| 2003      | Paralegal                                 | Raymond B. Schwartzberg & Associates                                            |
| 2001-2002 | Client Advocate                           | Student Health Outreach (SHOut)                                                 |
| 2000-2001 | Program Assistant                         | Yale International Affairs Council                                              |

## Major Administrative Leadership Positions

### Local

|           |                                                                                   |                |
|-----------|-----------------------------------------------------------------------------------|----------------|
| 2012-2019 | Network Manager, Brigham and Women's Primary Care Practice-Based Research Network | Brigham Health |
|-----------|-----------------------------------------------------------------------------------|----------------|

### Regional

|              |                                                                                                                                                          |                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 2025-present | Founder and General Chair, RAIIDIUS (Responsible Artificial Intelligence for Infectious Disease Intervention, Understanding, and Surveillance) Symposium | Columbia University/<br>American Sexually Transmitted Diseases Association (ASTDA) |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|

### National

|              |                                                                                     |                                                                        |
|--------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 2024-present | Planning Committee Member, Annual New England Science Symposium                     | Biomedical Science Careers Program                                     |
| 2024         | Co-organizer, Using Artificial Intelligence to Better Understand Menopause Workshop | National Science Foundation (NSF)/ National Institutes of Health (NIH) |

|                              |                                                                                                   |                                                      |
|------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 2018<br><i>International</i> | Co-organizer, Queer Health Hackathon                                                              | Tegan and Sara Foundation                            |
| 2026                         | Co-organizer/Registration Chair, Conference on Fairness, Accountability, and Transparency (FAccT) | Association for Computing Machinery (ACM)            |
| 2023                         | Co-organizer/Author Mentorship Program Sub-Chair, Machine Learning for Health (ML4H) Symposium    | Association for Health Learning and Inference (AHLI) |

## Committee Service

|              |                                                                                                                      |                              |
|--------------|----------------------------------------------------------------------------------------------------------------------|------------------------------|
| 2025–present | AI at VP&S Governance and Oversight Working Group                                                                    | Columbia University          |
| 2025–present | 2025–present                                                                                                         | Member                       |
| 2025–present | AI at VP&S Community Working Group                                                                                   | Columbia University          |
| 2025–present | 2025–present                                                                                                         | Member                       |
| 2021–2023    | Department of Biomedical Informatics Diversity, Equity, and Inclusion Working Group                                  | Columbia University          |
| 2021–2023    | 2021–2023                                                                                                            | Founding Member              |
| 2021–2022    | Toward Diversity, Equity, and Inclusion in Informatics, Health Care, and Society Seminar Series & Planning Committee | Columbia University          |
| 2021–2022    | 2021–2022                                                                                                            | Founding Member              |
| 2020–2022    | Biomedical Informatics Departmental Seminar Series Planning Committee                                                | Columbia University          |
| 2020–2022    | 2020–2022                                                                                                            | Member                       |
| 2019–2021    | Electronic Health Record Phenotyping Working Group                                                                   | Columbia University          |
| 2019–2021    | 2019–2021                                                                                                            | Member                       |
| 2015–2018    | Sexual Orientation and Gender Identity Working Group                                                                 | Partners Healthcare          |
| 2015–2018    | 2015–2018                                                                                                            | Founding Member              |
| 2015–2016    | Equity and Diversity Council                                                                                         | Brigham Health               |
| 2015–2016    | 2015–2016                                                                                                            | Founding Member              |
| 2015–2016    | 2015–2016                                                                                                            | Co-Chair, Data Working Group |
| 2013–2015    | Partners eCare Research Working Group                                                                                | Partners Healthcare          |
| 2013–2015    | 2013–2015                                                                                                            | Member                       |
| 2013–2015    | Partners eCare Sexual Orientation and Gender Identity Working Group                                                  | Partners Healthcare          |
| 2013–2015    | 2013–2015                                                                                                            | Founding Member              |
| 2013–2014    | Partners Human Research Committee, Insight Usability Working Group                                                   | Partners Healthcare          |
| 2013–2014    | 2013–2014                                                                                                            | Member                       |
| 2013–2014    | Partners Enterprise Research Portal Working Group                                                                    | Partners Healthcare          |
| 2013–2014    | 2013–2014                                                                                                            | Member                       |

|           |                                                             |                         |
|-----------|-------------------------------------------------------------|-------------------------|
| 2011–2019 | LGBTQ and Allies Employee Resource Group                    | Brigham Health          |
|           | 2014–2016                                                   | Co-Chair                |
|           | 2011–2019                                                   | Member                  |
| 2007–2008 | Association of Multicultural Members of Partners Healthcare | Partners Healthcare     |
|           | 2007–2008                                                   | Member                  |
|           | 2007–2008                                                   | Vice-Chair, BWH Chapter |

## Professional Societies

|              |                                                                                            |                                                  |
|--------------|--------------------------------------------------------------------------------------------|--------------------------------------------------|
| 2023-present | Society for the Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS) |                                                  |
| 2022-present | Association for Computational Linguistics (ACL)                                            |                                                  |
| 2022-present | Association for the Advancement of Artificial Intelligence (AAAI)                          |                                                  |
| 2022-present | Machine Learning for Healthcare (MLHC)                                                     |                                                  |
| 2020-present | Latinx in AI                                                                               |                                                  |
| 2019-present | Observational Health Data Science and Informatics (OHDSI) Collaborative                    |                                                  |
|              | 2024-present                                                                               | Co-lead, Early-Stage Researchers Workgroup       |
| 2019-present | New York Academy of Sciences                                                               |                                                  |
| 2017-present | Association for Computing Machinery (ACM)                                                  |                                                  |
| 2014-present | American Medical Informatics Association (AMIA)                                            |                                                  |
| 2012–2014    | American Public Health Association (APHA)                                                  |                                                  |
| 2012–2014    | AcademyHealth                                                                              |                                                  |
| 2008-present | Biomedical Science Careers Program (BSCP)                                                  |                                                  |
|              | 2023-present                                                                               | Member, Educational Relief Fund Review Committee |

## Grant Review Activities

|      |                                                                                         |     |
|------|-----------------------------------------------------------------------------------------|-----|
| 2026 | Grant Review Panelist, Directorate for Technology, Innovations, and Partnerships (TIPS) | NSF |
|------|-----------------------------------------------------------------------------------------|-----|

## Editorial Activities

### *Ad hoc Reviewer*

|                                      |                                                                  |
|--------------------------------------|------------------------------------------------------------------|
| American Psychologist                | Learning Health Systems                                          |
| Annals of Internal Medicine          | Journal of Medical Internet Research                             |
| Applied Clinical Informatics         | Journal of the American Medical Informatics Association          |
| Clinical Infectious Disease          | Journal of the International AIDS Society                        |
| Frontiers in Digital Health          | Medical Care                                                     |
| Frontiers in Medicine                | Natural Language Processing                                      |
| Frontiers in Reproductive Health     | npj Digital Medicine                                             |
| Health Informatics Journal           | npj Women's Health                                               |
| Health Services Research             | Open Forum Infectious Diseases                                   |
| JAMA Network Open                    | The Joint Commission Journal on Quality and Patient Safety       |
| JAMIA Open                           | The Journal of Infectious Diseases                               |
| Journal of Biomedical Informatics    | Transactions of the Royal Society of Tropical Medicine & Hygiene |
| Journal of General Internal Medicine |                                                                  |

*Other Editorial Roles*

2021–2023 Journal of the American Medical Informatics Association (JAMIA)  
 2022–2023, JAMIA Journal Club Manager  
 2021–2023, Student Editorial Board Member

**Honors and Prizes**

|           |                                                                    |                                                                                                                            |            |
|-----------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------|
| 2025      | STAT Wunderkind (Top-30 Early Career Researchers in North America) | STAT, Boston Globe Media                                                                                                   | Research   |
| 2025      | Distinguished Poster Award                                         | American Medical Informatics Association (AMIA)                                                                            | Research   |
| 2025      | OHDSI Titan Award Nominee                                          | Observational Health Data Science and Informatics (OHDSI) Collaborative                                                    | Leadership |
| 2024–2025 | AINet Fellow for Safety and Security in Artificial Intelligence    | Deutscher Akademischer Austauschdienst (DAAD) German Academic Exchange Service, Federal Ministry of Education and Research | Research   |
| 2022–2024 | T15 Predoctoral Research Training Fellowship                       | National Library of Medicine/ National Institute of Allergy and Infectious Diseases                                        |            |
| 2021–2025 | Computational and Data Science Fellowship                          | Association for Computing Machinery (ACM) Special Interest Group in High Performance Computing (SIGHPC)                    | Academic   |
| 2021–2023 | Hope Scholarship                                                   | Biomedical Science Careers Program                                                                                         |            |
| 2021–2022 | HSF Scholarship                                                    | Hispanic Scholarship Fund (HSF)                                                                                            |            |
| 2019–2022 | T15 Predoctoral Research Training Fellowship                       | National Library of Medicine                                                                                               |            |
| 2019–2020 | Dean's Fellowship                                                  | Columbia University Vagelos College of Physicians and Surgeons                                                             | Academic   |
| 2018      | Ford Foundation Predoctoral Fellowship, Honorable Mention          | National Academies of Sciences, Engineering, and Medicine                                                                  | Academic   |
| 2018      | Partners in Excellence Award for Leadership and Innovation         | Partners HealthCare                                                                                                        |            |
| 2018      | Ruth and William Silen MD Award, Third Prize                       | New England Science Symposium                                                                                              | Research   |
| 2017      | The Open Gate Travel Grant                                         | Harvard Gender & Sexuality Caucus                                                                                          | Research   |
| 2016      | LGBTQ Leadership Award                                             | Brigham Health                                                                                                             |            |
| 2015      | Brigham Way honoree                                                | Brigham and Women's Hospital                                                                                               | Leadership |
| 2015      | Travel Scholarship                                                 | North American Primary Care Research Group                                                                                 |            |
| 2014      | Partners in Excellence Award for Quality Treatment and Service     | Partners HealthCare                                                                                                        |            |
| 2014      | Outstanding Quality & Patient Safety Oral Presentation             | Society of General Internal Medicine                                                                                       | Research   |
| 2012      | Partners in Excellence Award for Leadership and Innovation         | Partners HealthCare                                                                                                        |            |
| 2004      | Mellon Forum Undergraduate Thesis Research Award                   | Yale University                                                                                                            |            |

## Report of Funded Projects

### *Funding Information*

#### **Current**

- 2026-2028    Evaluating the Fetal and Infant Safety of Doxycycline: A Global OHDSI Network Study  
ASTDA Summer Research Fellowship  
American Sexually Transmitted Diseases Association  
Principal Investigator  
Aim: To examine the association, if any, between prenatal exposure to doxycycline and increased risk of major congenital malformations (MCMs) compared with pregnancy-appropriate alternative antibiotics prescribed for the same infection.
- 2026–2028    The Empilisweni Initiative - Advancing Implementation of Cervical Cancer Control  
NCI 5U54CA284030  
Co-Investigator (PI: Castor)  
Aim: To use a web-based Single Patient Viewer to integrate and visualize longitudinal, cross-site data from South Africa in order to support evidence-based interventions for cervical cancer and its primary cause, human papillomavirus (HPV).
- 2025-2026    Symposium on Responsible Artificial Intelligence and Informatics for the Prevention and Control of Sexually Transmitted Infections  
Small Project Assistance Award  
American Sexually Transmitted Diseases Association  
Principal Investigator  
Aim: To convene AI researchers, clinicians, epidemiologists, public health specialists, bioethicists, and trainees for a one-day symposium at Columbia University Irving Medical Center to explore emerging responsible AI and informatics applications in diagnosis, surveillance, treatment, and prevention of sexually transmitted infections.
- 2025-2026    Spatiotemporal Mental Health-related Investigations to Foster Thriving among LGBTQ+, Black, and/or Latinx Youth in the United States (SHIFTS)  
Early Career Research Grant  
Hopelab  
Co-Principal Investigator  
Aim: To conduct population-level examinations of mental health outcomes and their associated factors in American LGBTQ+ and/or racially minoritized youth across 2018, 2021, and 2024 Hopelab National Surveys.
- 2024-2026    Development and Evaluation of an Informatics-Based Approach to Reducing Disparities in Kidney Transplant Evaluation Completion in a Learning Health System  
Learning Health System and Health Justice Research and Training Award  
EQUIP Center for Learning Health Systems/Dalio Center for Health Justice  
Co-Investigator (PI: H. Fernandez)  
Aim: To identify patients at high risk for delayed kidney pre-transplant clinical evaluation using machine learning, target them for enhanced monitoring and engagement, and provide direct support to mitigate roadblocks contributing to delays.

**Past**

- 2024 Deep probabilistic predictive models for stroke and coronary heart disease  
NHLBI R01HL148248  
Graduate Research Student (Site PI: Elhadad)  
Aim: To construct and evaluate deep learning methods for coronary heart disease and stroke risk assessment from electronic health records and develop techniques to incorporate clinical text, handle missing data, and evaluate fairness of deep learning for cardiovascular risk scores.
- 2023-2024 Clinical and Translational Science Award  
National Center for Advancing Translational Sciences (NCATS) 3UL1TR001873-08S1  
Graduate Research Student (PI: Reilly)  
Aim: To investigate justice, equity, diversity and inclusion (JEDI) in ethical frameworks applied to research on artificial intelligence in digital health and create educational case studies for identifying ethical issues related to JEDI and discussing potential approaches to integrate JEDI goals.
- 2018-2019 A Multicenter Randomized Controlled Trial of Pharmacist-Directed Transitional Care to Reduce Post-Hospitalization Utilization  
National Institute on Aging (NIA) R01 AG 058911-01  
Project Manager (Site PI: Schnipper)  
Aim: To measure the impact of pharmacist-led discharge (PHARM-DC) interventions on post-discharge utilization within 30 days and identify patient sub-populations most likely to benefit from PHARM-DC.
- 2017-2018 Novel electronic health record phenotyping of LGBTQ intersectional identities and associated health disparities using natural language processing and machine learning  
Harvard Catalyst  
Co-Investigator (PI: Zhou)  
Aim: To develop methods using machine learning and natural language processing to identify sexual and gender minority (SGM) populations from electronic health records and characterize potential healthcare disparities associated with different segments of SGM populations.
- 2016-2018 Electronic Medication Adherence Reporting and Feedback During Care Transitions (Smart Pillbox)  
Agency for Healthcare Research and Quality (AHRQ) R21 HS 024587-01  
Project Manager/Statistical Programmer (PI: Schnipper)  
Aim: To implement a smart pillbox intervention for patients discharged from the hospital to the community and evaluate the effects of the intervention on post-discharge medication discrepancies, medication adherence, and chronic disease management.
- 2015-2018 Implementation of a Medication Reconciliation Protocol to Improve Patient Safety  
Agency for Healthcare Research and Quality (AHRQ) R18 HS 023757-01A1  
Project Manager/Statistical Programmer (PI: Schnipper)  
Aim: To widely disseminate, implement, and evaluate sustainable medication reconciliation interventions that improve patient safety during care transitions.
- 2015-2017 Clinician-Patient Relationships: Boundaries, Barriers, Breakdowns  
Gold Foundation/Lucian Leape Family Foundation  
Project Director (PI: Schiff)  
Aim: To examine physician attitudes and practices related to caring acts that have been questioned as “inappropriate” or “unethical” crossing of professional boundaries.

- 2014–2016 Understanding and Preventing Diagnostic Pitfalls  
Risk Management Foundations of the Harvard Medical Institutions (CRICO)  
Project Director (PI: Schiff)  
Aim: To identify leading diagnostic pitfalls that pose safety and malpractice risks, then design and test the accuracy of electronic screens for these pitfalls in the electronic health record.
- 2014–2016 Evaluating Variation in Overactive Bladder Medication Use and Other Treatments  
Astellas Pharma US, Inc.  
Statistical Programmer (PI: Loughlin)  
Aim: To develop an electronic phenotype for overactive bladder, then characterize overactive bladder treatment, medication adherence, and outcomes.
- 2014–2015 Are Physicians and Patients Adhering to the Published Guidelines for Fasting during Ramadan?  
Martin P. Solomon Award for Primary Care Scholarship, BWH  
Co-Investigator (PI: Ali)  
Aim: To measure differences in rates of health care utilization for Muslim patients just prior to, during, and after Ramadan.
- 2014 First Annual BWH DGIM Research Day  
BRI/ORC Microgrant Pilot Program  
Principal Investigator  
Aim: To develop a BWH Division of General Internal Medicine and Primary Care retreat to showcase contributions of research assistants and promote their career development.
- 2013–2015 Patient Outcomes Associated with Year-End Departure of Resident Primary Care Physicians  
Support for Excellence in Educational Development (SEED) Grant, BWH  
Co-Investigator (PI: Solomon)  
Aim: To investigate whether acute care utilization is increased disproportionately among patients of graduating resident physicians compared to non-graduating residents.
- 2013–2014 Acute Respiratory Infection Telephone Management Service  
Brigham and Women's Physicians Organization Care Redesign Incubator and Startup Program – Round 1 and Round 2  
Data Manager (PI: Linder)  
Aim: To develop, implement, and evaluate an acute respiratory infection telephone management service at one of the Brigham and Women's Primary Care Practices.
- 2012–2014 Use of Behavioral Economics to Improve Treatment of Acute Respiratory Infections  
National Institutes of Health/University of Southern California 1RC4AG039115-01  
Data Manager/Statistical Programmer (Site PI: Linder)  
Aim: To evaluate the use of behavioral economic concepts such as enhanced defaults, alternative options, and social norms to reduce inappropriate antibiotic prescribing for acute respiratory infections.
- 2012–2014 Medical Malpractice and Patient Safety Protocol (PROMISES)  
Agency for Healthcare Research and Quality (AHRQ) 1R18HS019508-01  
Project Manager (PI: Schiff)  
Aim: To test the impact of quality improvement techniques to improve 3 high risk ambulatory malpractice areas: 1) medication management, 2) test ordering and results management 3) follow-up and referral management.

- 2008–2012 President’s Emergency Plan for AIDS Relief (PEPFAR)  
Health Resources and Services Administration (HRSA) U51HA02522  
Data Quality Assurance Manager / Data, Training, & Education Coordinator (PI: Kanki)  
A \$362 million grant to train health care workers, develop monitoring and evaluation systems, strengthen health care infrastructures, and collaborate with local hospitals and clinics in Nigeria, Botswana, and Tanzania that provide treatment for people living with HIV/AIDS.
- 2007–2008 Pharmacist Intervention to Decrease Medication Errors in Heart Disease Patients  
National Heart, Lung, and Blood Institute (NHLBI), R01HL089755  
Research Assistant (Site PI: Schnipper)  
Aim: To evaluate a literacy-focused program that provides pharmacist-led educational assistance at the time of hospital discharge to patients with heart disease.
- 2007 Influenza in the Partners Primary Care Practice-Based Research Network  
Roche Laboratories, RLI 06001207  
Research Assistant (PI: Linder)  
Aim: To measure and evaluate the appropriateness of antiviral and antibiotic prescribing for patients being treated for influenza in primary care clinics.

## Report of Local Teaching and Training

### *Teaching of Students in Courses*

|            |                                                                                                                              |                                                                                                 |
|------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 2026       | BINF G4001/COMS W4560 Introduction to Computational Biomedicine and Health<br>Course Instructor<br>Master’s and PhD students | Columbia University Graduate School of Arts and Sciences<br>2-hr sessions per week for 13 weeks |
| 2022       | BINF G4008 Interrogating Ethics and Justice in Digital Health<br>Teaching Assistant<br>Master’s and PhD students             | Columbia University Graduate School of Arts and Sciences<br>2-hr sessions per week for 13 weeks |
| 2021, 2022 | INTC M7210 Mechanisms and Practice Preceptor in Clinical Informatics<br>2nd year medical students                            | Columbia Vagelos College of Physicians and Surgeons<br>Two 2-hr sessions per week for 2 weeks   |
| 2020       | BINF G4011 Acculturation to Medicine & Clinical Informatics<br>Teaching Assistant<br>Master’s and PhD students               | Columbia University Graduate School of Arts and Sciences<br>2-hr sessions per week for 13 weeks |
| 2016, 2017 | HPM 299 Research with Large Databases<br>Teaching Fellow<br>Master’s and PhD students                                        | Harvard T.H. Chan School of Public Health<br>Five 1.5-hr sessions per week for 3 weeks          |

### *Mentored Trainees*

- 2025–2026 Cameron Marsh, Master of Public Health Candidate  
Second Reader, Thesis Committee, Columbia Mailman School of Public Health
- 2024 Kemka Ihemelandu, Undergraduate at Georgetown University  
Summer Mentor, Summer Program for Undergraduate Rising Stars (SPURS)

*Local Invited Presentations*

No presentations below were sponsored by 3rd parties/outside entities.

- 2025 Foundation Models Across Scales: From Cells to Health Systems / Workshop Moderator  
AI at Vagelos College of Physicians and Surgeons (AI@VP&S) Initiative, New York, NY
- 2024 The Promise and Perils of Artificial Intelligence for Health Equity / Invited Presentation  
Artificial Intelligence Summer Course, Columbia University Double Discovery Program.
- 2024 Elucidating Health Inequities and Research Gaps in HIV and Other Sexually Transmitted  
Infections Using Data Mining and a Large Language Model: A Computational Scoping Review /  
Invited Presentation  
HIV Prevention Research Working Group, NYPH, New York, NY.
- 2023 Development and Validation of Multimodal Deep Learning Models to Identify Human  
Monkeypox in the Electronic Health Record / Invited Presentation  
HIV Prevention Research Working Group, NYPH, New York, NY.
- 2023 Health Information Exchange Enables Enhanced STI Surveillance Using Electronic Health  
Record Data / Invited Presentation  
HIV Prevention Research Working Group, NYPH, New York, NY.
- 2023 Enabling a Learning Public Health System: Enhanced Surveillance of Sexually Transmitted  
Infections / Invited Presentation  
Division of Infectious Diseases, NewYork-Presbyterian Hospital (NYPH), New York, NY.
- 2022 Mining the Health Disparities and Minority Health Bibliome / Invited Presentation  
Biomedical Informatics Seminar Series, Department of Biomedical Informatics, Columbia  
University
- 2019 LGBTQ Health - Diversity, Culture, and Inclusion Series / Invited Presentation  
Division of General Internal Medicine and Primary Care, Brigham and Women's Hospital
- 2018 Documentation of Sexual Orientation and Gender Identity in the Electronic Health Record /  
Invited Presentation  
Division of General Internal Medicine and Primary Care, Brigham and Women's Hospital
- 2017 Improving and Leveraging Sexual Orientation and Gender Identity Demographics in Epic /  
Invited Presentation  
Sexual Orientation and Gender Identity Working Group, Partners HealthCare
- 2017 Algorithmic Bias and Health Disparities / Invited Presentation  
Machine Learning Working Group, Partners HealthCare
- 2014 The LGBTQ Data Collection Gap: Sexual Orientation and Gender Identity in the EHR / Invited  
Presentation  
LGBTQ and Allies Employee Resource Group, Brigham Health
- 2014 Designing and Building Clinical Research Databases / Invited Presentation  
Division of General Internal Medicine and Primary Care, Brigham and Women's Hospital
- 2014 SQL-based Data Management and Manipulation / Invited Presentation  
Division of General Internal Medicine and Primary Care, Brigham and Women's Hospital
- 2013 How Busy Clinicians Can Improve Care and Pursue Scholarship / Invited Workshop  
Division of General Internal Medicine and Primary Care, Brigham and Women's Hospital

## Report of Regional, National, and International Invited Teaching and Presentations

No presentations below were sponsored by 3rd parties/outside entities.

### *Regional*

- 2026 Development of Machine Learning Models to Identify Patients at High Risk for Delays in Kidney Transplant Evaluation Completion in a Learning Health System / Invited Presentation  
Icahn School of Medicine at Mount Sinai, Artificial Intelligence and Emerging Technology (AIET) Seminar Series  
New York, NY
- 2023 Getting into Graduate School and Succeeding/ Invited Panelist  
New England Science Symposium, Biomedical Science Careers Program  
Boston, MA
- 2022 Mining the Health Disparities and Minority Health Bibliome / Invited Presentation  
Biomedical Informatics Colloquium, New York City College of Technology (City Tech)  
New York, NY
- 2018 High physician and clinic-level variation in documentation of sexual orientation and gender identity in the electronic health record / Plenary Session  
New England Science Symposium  
Boston, MA  
**Received the Ruth and William Silen Award, Third Prize**
- 2016 Identifying and Understanding Malpractice-Prone Diagnostic Pitfalls / Invited Presentation  
Risk Management Foundation of the Harvard Medical Institutions Incorporated (RMF)  
CRICO Research Day  
Boston, MA

### *National*

- 2026 Toward Precision Population Health for Infectious Disease Control and Prevention in a Learning Health System: Lessons from the ARMOR Household Study (Selected Oral Abstract)  
American Medical Informatics Association Annual Symposium  
Dallas, TX
- 2026 Building a Learning Public Health System: Responsible AI and Informatics for Infectious Disease Prevention and Control (Invited Presentation)  
University of Pittsburgh, Department of Biomedical Informatics Colloquium  
Pittsburgh, PA
- 2026 Toward Precision Population Health for Infectious Disease Prevention and Control in a Learning Health System (Invited Presentation)  
University of Minnesota, Center for Learning Health System Sciences (CLHSS) iImpact Seminar Series  
Minneapolis, MN

- 2026 Building a Responsible AI Community for Infectious Diseases (Plenary Presentation)  
Responsible AI for Infectious Disease Intervention, Understanding, and Surveillance (RAIDIUS)  
Symposium  
New York, NY
- 2023 Enabling a Learning Public Health System: Enhanced Surveillance of Sexually Transmitted  
Infections (Selected Oral Abstract)  
American Medical Informatics Association Annual Symposium  
New Orleans, LA
- 2023 Challenges, Opportunities, and Considerations: Promoting Inclusive Research in the Era of Big  
Data / Invited Presentation and Panelist  
American Medical Informatics Association Annual Symposium  
New Orleans, LA
- 2022 Mining the Health Disparities and Minority Health Bibliome: A Computational Scoping Review  
(Selected Oral Abstract)  
American Medical Informatics Association Annual Symposium  
Washington, DC  
**Spotlighted by AMIA for “demonstrating best practices in promoting diversity, equity, and  
inclusion through scholarly communications in biomedical informatics.”**
- 2021 Differential Presentation and Delays in Treatment for Acute Myocardial Infarction Associated  
with Sex, Race, Ethnicity (Selected Oral Abstract)  
American Medical Informatics Association Annual Symposium  
San Diego, CA
- 2015 Advocacy and Implementation: Gathering SOGI Demographics in the Clinical Setting  
(Selected Oral Abstract)  
Gay and Lesbian Medical Association Annual Conference  
Portland, OR
- 2014 Comparing Clinicians’ Perception of Their Own and Their Peers’ Antibiotic Prescribing to  
Actual Antibiotic Prescribing for Acute Respiratory Infections in Primary Care (Selected Oral  
Abstract)  
Society of General Internal Medicine Annual Meeting  
San Diego, CA  
**Received the Outstanding Quality and Patient Safety Oral Presentation Award**

### *International*

- 2025 Research Roundtable on Population Health and Survival Analysis / Senior Chair  
Machine Learning for Health (ML4H) Symposium  
San Diego, CA, USA
- 2025 Plenary Session Lightning Talks / Moderator  
OHDSI Global Symposium  
New Brunswick, NJ, USA

- 2023 Health Information Exchange Enables Enhanced STI Surveillance Using Electronic Health Record Data (Selected Oral Abstract)  
STI & HIV 2023 World Congress  
Chicago, IL, USA
- 2015 Gathering Sexual Orientation and Gender Identity Demographics in the Clinical Setting / Selected Workshop  
North American Primary Care Research Group Practice-Based Research Network Conference  
Bethesda, MD, USA
- 2014 PBRN Organization and Governance to Promote Practice, Clinician, Researcher, and Patient Engagement / Selected Workshop  
North American Primary Care Research Group Practice-Based Research Network Conference  
Bethesda, MD, USA
- 2011 Programming: Creating a Utility and Basics of Scripting/ Invited Seminar  
AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria
- 2011 Programming and Trouble-shooting Databases/ Invited Seminar  
AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria
- 2011 Cohort Analyses II: Using EHR Data/ Invited Seminar  
AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria
- 2011 Cohort Analyses I: Using EHR Data/ Invited Seminar  
AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria
- 2011 Monitoring data and Trouble-shooting/ Invited Seminar  
AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria
- 2011 Introduction to Data Management/ Invited Seminar  
AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria
- 2011 Critical Elements in Database Cleaning, a Checklist/ Invited Seminar  
AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria
- 2011 Basics of Programming and Scripting/ Invited Seminar  
AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria
- 2009 Improving Understanding and Medication Adherence/ Invited Plenary Presentation  
HIV Training, AIDS Prevention Initiative Nigeria Plus (APIN Plus)  
Abuja, Nigeria

## Report of Clinical Activities and Innovations

### *Past and Current Licensure and Board Certification*

2006 Certification, New York State Emergency Medical Technician (EMT-D)

## Report of Teaching and Education Innovations

Creation of a JEDI-AI  
Case Study Series  
(2023-2024)

Co-developed educational case studies focused on justice, ethics, diversity, and inclusion (JEDI) in the development and use of artificial intelligence (AI) approaches in digital health.

## Report of Technological and Other Scientific Innovations

ESCOBA: End-to-end Supervised  
Clustering of Outcomes for Binary  
Assessment  
(2025)

ESCOBA is a novel machine learning-based subgroup discovery approach with broad applications for advancing precision population health. ESCOBA was created to better inform efforts to recognize and engage subgroups who might benefit from greater attention and more closely tailored clinical and public health interventions.

HDMH Monitor  
(2022)

HDMH Monitor is an online article repository and interactive dashboard that leverages natural language processing and machine learning methods to support scientific discovery via automated archive, search, information synthesis, and data visualization of articles concerning health disparities and minority health (HDMH).  
<https://hdmhmonitor.dbmi.columbia.edu/>

C-REACT: Contextualized  
Race and Ethnicity  
Annotations for Clinical Text  
(2022)

As a member of the Elhadad lab, my colleagues and I curated the C-REACT dataset, a large publicly-available corpus of sentences from clinical notes manually annotated for information related to race and ethnicity (RE) This corpus contains two sets of gold-standard annotations for RE data. The first set contains granular RE-information such as patient country of origin. The second set of annotations contains RE labels manually assigned by clinicians.  
<https://physionet.org/content/race-ethnicity-clinical-text/>

Health Equity Research  
Assessment (HERA)  
(2021)

As a member of the Elhadad lab, I supported the Health Equity Research Assessment (HERA), a large-scale characterization conducted across Observational Health Data Science Informatics (OHDSI) sites with heterogeneous populations and insurance coverage, allowing for identification of persistent and generalizable trends in diagnosis differences. The HERA dashboard and visualizations can be used to download study data, further investigate health differences, and generate novel hypotheses.  
<https://data.ohdsi.org/HERACharacterization/>

|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CovidWatcher<br>(2020)                                                  | As part of a department-wide effort, my colleagues and I developed CovidWatcher, an app and online portal that surveyed users about COVID-19 exposure, symptoms, access to medical care, and impact on daily life. Data collected was used to track spread of COVID-19, giving citizens real-time information about hot spots and enabling health care officials to deploy resources where needed. <a href="https://www.dbmi.columbia.edu/covidwatcher/">https://www.dbmi.columbia.edu/covidwatcher/</a>         |
| EHR pediatric quality improvement tool<br>(2011)                        | Co-developed and evaluated a utility to extract information from data warehouse and generate measures based on 23 pediatric quality of care indicators at 33 Harvard PEPFAR sites in Nigeria. Programmed module to review continuity of care, drug therapy initiation, medication adherence, loss to follow-up, laboratory monitoring, disease screening based on clinical symptoms assessment or diagnostic evaluation, treatment failure, toxicity management, nutritional assessment, and treatment response. |
| EHR data quality assessment tool, version 2.0<br>(2011)                 | Instituted major modifications to and evaluated a pre-existing EHR utility to incorporate expanded data quality indicators at 33 Harvard PEPFAR sites in Nigeria. With this utility, central and site-level data managers are able to better monitor and improve data completeness, validity, accuracy, and currency.                                                                                                                                                                                            |
| Electronic health record (EHR) adult quality improvement tool<br>(2010) | Co-developed and evaluated a utility to extract information from data warehouse and generate measures based on 15 adult quality of care indicators at 33 Harvard PEPFAR sites in Nigeria. Module reviews continuity of care, drug therapy initiation, loss to follow-up, laboratory monitoring, disease screening based on clinical symptoms assessment, treatment failure, and treatment response.                                                                                                              |

## Report of Education of Patients and Service to the Community

### *Activities*

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020-present | De La Salle Academy/Proctor<br>Proctor annual entrance exam for only private, independent middle school in New York City for academically talented students from economically less advantaged families.                                                                                                                                                                                                                                                                                           |
| 2019         | High School Scholars Program, American Medical Informatics Association/Mentor<br>Supported high school students with research projects; reviewed their work and helped them practice scientific presentations; offered career advice and encouragement.                                                                                                                                                                                                                                           |
| 2013-2016    | Human Rights Commission, City of Cambridge/Commissioner, Vice Chair<br>Adjudicated complaints of discrimination occurring in Cambridge, MA related to housing, employment, education, and public accommodation. Lobbied the state legislature on important policies such as pay equity, protections relative to domestic violence, and anti-discrimination legislation for sexual and gender minorities. Facilitated conciliations among parties. Conducted information sessions on civil rights. |

- 2007 Timmy Global Health, Dominican Republic/Medical Relief Volunteer  
Helped establish temporary medical clinics along Haiti-Dominican Republic border; triaged patients, managed pharmacy, translated proper drug use and application.
- 2006-2016 Boston Medical Reserve Corps/Emergency Medical Technician  
The Boston Medical Reserve Corps is a group of volunteers who are ready to respond should there be a public health emergency in Boston.
- 2001-2005 Latino Youth Enrichment Team (LYET), New Haven, CT/Co-Director  
Managed community center mentorship program and developed instructional methods for Latino youth ranging in age from 6 through 13.

### *Educational Material for Patients and the Lay Community*

#### **Books, monographs, articles and presentations in other media**

- |      |                                                                                                                                                                |           |                                        |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------|
| 2012 | Harvard School of Public Health/AIDS Prevention Initiative in Nigeria: Reducing the Burden of AIDS U.S. President's Emergency Plan for AIDS Relief             | Co-author | Harvard PEPFAR Nigeria Program Report  |
| 2012 | Harvard School of Public Health/Management and Development for Health: Reducing the Burden of AIDS in Tanzania U.S. President's Emergency Plan for AIDS Relief | Co-author | Harvard PEPFAR Tanzania Program Report |
| 2012 | Botswana-Harvard AIDS Institute Partnership: Reducing the Burden of AIDS U.S. President's Emergency Plan for AIDS Relief                                       | Co-author | Harvard PEPFAR Botswana Program Report |

### **Recognition**

- |      |                                                                                                                                                                                       |                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 2025 | Feature article "Harry Reyes Nieva: Human-Centered AI for Health Care" discussing my background and research                                                                          | CUIMC News                |
| 2025 | Interviewed as a 2025 STAT Wunderkind, one of the top 30 early career scientists in North America                                                                                     | STAT News                 |
| 2019 | Spotlighted in the article "On the Margins" about research using natural language processing to study LGBTQ health                                                                    | Harvard Medicine Magazine |
| 2019 | Interviewed for "Machine Learning and Health Equity" episode to discuss leveraging AI in health equity research                                                                       | ThinkResearch Podcast     |
| 2015 | Publicity for successful advocacy leading to routine collection of sexual orientation and gender identity demographics in the Mass General Brigham (formerly Partners Healthcare) EHR | Boston Magazine           |

## Report of Scholarship

*Peer reviewed publications in print or other media*

### Research Investigations

1. Zucker J, Huang S, Schrikkut D, Flanagan M, Li J, Ramirez J, Perez E, Theodore DA, **Reyes Nieva H**, Heck CJ, Carnevale C, Meyers K, Borrell LN, Uhlemann AC, Sobieszczyk ME, Castor D. Stable Does it: User Patterns and Determinants of Doxycycline-PEP Use in Real World Settings. *JAMA Network Open*. In press.
2. Tucker E, **Reyes Nieva H**, Castor D, Yin MT, Gordon P, Elhadad N, Zucker J. Trends in Triple-Site Testing for Chlamydia and Gonorrhea in New York City: A Cross-sectional Study Using Health Information Exchange Data. *Sexually Transmitted Diseases*. 2026 Mar 19. Epub ahead of print. PMID:41854257. doi:10.1097/OLQ.0000000000002301.
3. **Reyes Nieva H**, Zucker J, Tucker EG, McLean J, DeLaurentis C, Gunaratne S, Elhadad N. Development of Machine Learning-Based Mpox Surveillance Models in a Learning Health System. *Sexually Transmitted Infections*. 2025 Oct 16;101(7):456-460. doi:10.1136/sextrans-2024-056382.
4. Ketenci M, Jeanselme V, **Reyes Nieva H**, Joshi S, Elhadad N. ADHAM: Additive Deep Hazard Analysis Mixtures for Interpretable Survival Regression. *Machine Learning for Healthcare (MLHC)*; 15 Aug 2025. doi: 10.48550/arXiv.2509.07108.
5. **Reyes Nieva H**, Zucker J, Tucker EG, Castor D, Yin M, Gordon P, Elhadad N. Enhanced Surveillance of Sexually Transmitted Infections to Foster a Learning Public Health System. *JAMA Netw Open*. 2025 Jun 2;8(6):e2514308. PMID: 40526386. doi:10.1001/jamanetworkopen.2025.14308.
6. Bear Don't Walk OJ IV, Pichon A, **Reyes Nieva H**, Sun T, Li J, Joseph J, Kinberg S, Richter LR, Crusco S, Kulas K, Ahmed SA, Snyder D, Rahbari A, Ranard BL, Juneja P, Demner-Fushman D, Elhadad N. Contextualized race and ethnicity annotations for clinical text from MIMIC-III. *Scientific Data*. 2024 Dec 5;11(1):1332. PMID:39638783. doi:10.1038/s41597-024-04183-2.
7. **Reyes Nieva H**, Bakken S, Elhadad N. Mining the Health Disparities and Minority Health Bibliome: A Computational Scoping Review and Gap Analysis of 200,000+ Articles. *Science Advances*. 2024 Jan 26;10(4):eadf9033. PMID:38266089; PMCID:PMC10807818. doi:10.1126/sciadv.adf9033.
8. Bear Don't Walk IV OJ, Pichon A, **Reyes Nieva H**, Sun T, Altosaar J, Natarajan K, Perotte A, Tarczy-Hornoch P, Demner-Fushman D, Elhadad N. Auditing Learned Associations in Deep Learning Approaches to Extract Race and Ethnicity from Clinical Text. *AMIA Annu Symp Proc*. 2024 Jan 11;2023:289-298. PMID:38222422; PMCID:PMC10785932.
9. Tucker EG, **Reyes Nieva H**, Schiffer K, Yin MT, Castor D, Gordon P, Elhadad N, Zucker J. Using a Health Information Exchange to Characterize Changes in HIV Viral Load Suppression and Disparities During the COVID-19 Pandemic in NYC. *Open Forum Infectious Diseases*. 2023 Nov 29;10(12):ofad584. PMID:38156044; PMCID:PMC10754646. doi:10.1093/ofid/ofad584.
10. Spotnitz M, Acharya N, Cimino JJ, Murphy S, Namjou B, Crimmins N, Walunas T, Liu C, Crosslin D, Benoit B, Rosenthal E, Pacheco J, Ostroplets A, **Reyes Nieva H**, Patterson JS, Richter LR, Callahan TJ, Elhussein A, Pang C, Kiryluk K, Nestor J, Khan A, Mohan S, Minty E, Chung W, Wei WQ, Natarajan K, Weng C. A metadata framework for electronic phenotypes. *JAMIA Open*. 2023 Jul; 6(2):ooad032. PMID:37181728; PMCID:PMC10168627. doi:10.1093/jamiaopen/oad032.
11. Schnipper JL, **Reyes Nieva H**, Yoon C, Mallouk M, Mixon AS, Rennke S, Chu ES, Mueller SK, Smith GR, Williams MV, Wetterneck TB, Stein JM, Dalal A, Labonville S, Sridharan A, Stoldorf D, Orav J, Levin B, Gresham M, Goldstein J, Platt S, Nyenpan C, Howell E, Kripalani S, and MARQUIS2

Site Leaders for the MARQUIS2 Study Group. What Works in Medication Reconciliation: An On-Treatment and Site Analysis of the MARQUIS2 Study. *BMJ Qual Saf.* 2023 Aug;32(8):457-469. PMID:36948542. doi:10.1136/bmjqs-2022-014806.

**Selected as #1 among the "Top Articles of 2023" by BMJ Quality and Safety**

12. Chaplin B, Agbaji O, **Reyes Nieva H**, Olatunde B, Chang C, Mitruka K, Sule H, Dajel T, Zee A, Ahmed ML, Ahmed I, Okonkwo P, Rawizza H, Kanki P. The Timeliness of Point of Care Viral Load Results Improves HIV Monitoring in Nigeria. *Clin. Infect. Dis.* 2023 Feb 8;76(3):e671-e680. PMID:35872644. doi:10.1093/cid/ciac609.
13. Schnipper JL, **Reyes Nieva H**, Mallouk M, Mixon AS, Rennke S, Chu ES, Mueller SK, Smith GR, Williams MV, Wetterneck TB, Stein JM, Dalal A, Labonville S, Sridharan A, Stollendorf D, Orav J, Levin B, Gresham M, Yoon C, Goldstein J, Platt S, Nyenpan C, Howell E, Kripalani S, and MARQUIS2 Site Leaders for the MARQUIS2 Study Group. Effects of a Refined Evidence-based Toolkit and Mentored Implementation on Medication Reconciliation Quality and Safety at Eighteen Hospitals: Results of the MARQUIS2 Quality Improvement Study. *BMJ Qual Saf.* 2022 Apr;31(4):278-286. PMID:33927025. doi:10.1136/bmjqs-2020-012709.
14. Groisser A, **Reyes Nieva H**, Wright A, Ruan E, Schiff GD. Terminations in Primary Care: A Retrospective Observational Study of 16 Primary Care Clinics. *J Gen Intern Med.* 2022;37(3):548-555. PMID:33948801; PMCID:PMC8858375. doi:10.1007/s11606-021-06793-7.
15. Schiff GD, Volodarskaya M, Ruan E, Lim A, Wright A, Singh H, **Reyes Nieva H**. Characteristics of Disease-Specific and Generic Diagnostic Pitfalls: A Qualitative Study. *JAMA Netw Open.* 2022;5(1):e2144531. PMID:35061037; PMCID:PMC8783262. doi:10.1001/jamanetworkopen.2021.44531.
16. Bear Don't Walk OJ IV, **Reyes Nieva H**, Lee SSJ, Elhadad N. A Scoping Review of Ethics Considerations in Clinical Natural Language Processing. *JAMIA Open.* 2022;5(2):ooac039. PMID:35663112; PMCID:PMC9154253. doi:10.1093/jamiaopen/ooac039.
17. Shahani A, **Reyes Nieva H**, Czado K, Shannon E, Gaetani R, Gresham M, Cruz Garcia J, Ganesan H, Cerciello E, Dave J, Jain R, Schnipper JL. An Electronic Pillbox Intervention Designed to Improve Medication Safety During Care Transitions: Challenges and Lessons Learned regarding Implementation and Evaluation. *BMC Health Serv. Res.* 2022;22(1):1304. PMID:36309744; PMCID:PMC9618185. doi:10.1186/s12913-022-08702-y.
18. Yeh GY, Litrownik D, Wayne PM, Beach D, Klings ES, **Reyes Nieva H**, Pinheiro A, Davis RB, Moy ML. The BEAM Study (Breathing, Education, Awareness, Movement): a Randomized Controlled Feasibility Trial of Tai Chi Exercise in Patients with COPD. *BMJ Open Respiratory Research.* 2021;18(3):288-298. PMID:33219007; PMCID:PMC7682460. doi:10.1136/bmjresp-2020-000697.
19. Kraemer KM, Litrownik D, Moy ML, Wayne PM, Beach D, Klings ES, **Reyes Nieva H**, Pinheiro A, Davis RB, Yeh GY. Exploring tai chi exercise and mind-body breathing in patients with COPD in a randomized controlled feasibility trial. *COPD.* 2021;18(3):288-298. PMID:34106027; PMCID:PMC8283813. doi:10.1080/15412555.2021.1928037.
20. Linder JA, Weissman JS, **Reyes Nieva H**, Lipsitz S, Haring RS, DeAngelis J, Kristy RM, Loughlin KR. Overactive bladder in an integrated delivery system: a longitudinal cohort study. *BMC Health Serv Res.* 2020;20(1):447. PMID:32434511; PMCID:PMC7238545. doi:10.1186/s12913-020-05315-1.
21. **Reyes Nieva H**, Ruan E, Schiff G. Professional-Patient Boundaries: a National Survey of Primary Care Physicians' Attitudes and Practices. *J Gen Intern Med.* 2020;35(2):457-464. PMID:31755012; PMCID:PMC7018879. doi:10.1007/s11606-019-05543-0.

22. Sun TY, Bear Don't Walk OJ IV, Chen J, **Reyes Nieva H**, Elhadad N. Exploring Gender Disparities in Time to Diagnosis. Machine Learning for Health (ML4H) at NeurIPS 2020; 12/2020. arXiv:2011.06100.
23. Mixon AS, Smith GR, Mallouk M, **Reyes Nieva H**, Kripalani S, Rennke S, Chu E, Sridharan A, Dalal A, Mueller S, Williams M, Wetterneck T, Stein JM, Stollendorf D, Howell E, Orav J, Labonville S, Levin B, Yoon C, Gresham M, Goldstein J, Platt S, Nyenpan C, Schnipper JL. Design of MARQUIS2: study protocol for a mentored implementation study of an evidence-based toolkit to improve patient safety through medication reconciliation. *BMC Health Serv Res*. 2019 Sep 11; 19(1):659. PMID:31511070. PMCID:PMC6737715 doi:10.1186/s12913-019-4491-5.
24. Ali NB, **Reyes Nieva H**, Percac-Lima S, Shields HM, Linder JA, Yusuf Osman N. Health Care Utilization by Muslim Patients During Ramadan. *J Health Care Poor Underserved*. 2019;30(4):1360-1372. PMID:31680102. doi:10.1353/hpu.2019.0099.
25. Schiff GD, **Reyes Nieva H**, Griswold P, Leydon N, Ling J, Federico F, Keohane C, Ellis BR, Foskett C, Orav EJ, Yoon C, Goldmann D, Weissman JS, Bates DW, Biondolillo M, Singer SJ. Randomized Trial of Reducing Ambulatory Malpractice and Safety Risk: Results of the Massachusetts PROMISES Project. *Med Care*. 2017 08; 55(8):797-805. PMID:28650922; doi:10.1097/MLR.0000000000000759.
26. Schiff GD, **Reyes Nieva H**, Griswold P, Leydon N, Ling J, Biondolillo M, Singer SJ. Addressing Ambulatory Safety and Malpractice: The Massachusetts PROMISES Project. *Health Serv Res*. 2016 12; 51 Suppl 3:2634-2641. PMID:27892623; PMCID:PMC5134349. doi:10.1111/1475-6773.12621.
27. Kroll DS, **Reyes Nieva H**, Barsky AJ, Linder JA. Benzodiazepines are Prescribed More Frequently to Patients Already at Risk for Benzodiazepine-Related Adverse Events in Primary Care. *J Gen Intern Med*. 2016 09; 31(9):1027-34. PMID:27177914; PMCID:PMC4978684. doi:10.1007/s11606-016-3740-0.
28. Solomon SR, Gooding HC, **Reyes Nieva H**, Linder JA. Acute Care Utilization by Patients After Graduation of Their Resident Primary Care Physicians. *J Gen Intern Med*. 2015 Nov; 30(11):1611-7. PMID:25896089. PMCID:PMC4617921. doi:10.1007/s11606-015-3305-7.
29. Singer SJ, **Reyes Nieva H**, Brede N, Ling J, Leydon N, Weissman JS, Goldmann D, Griswold P, Yoon C, Orav EJ, Bates DW, Biondolillo M, Schiff GD. Evaluating ambulatory practice safety: the PROMISES project administrators and practice staff surveys. *Med Care*. 2015 Feb; 53(2):141-52. PMID:25464161. doi:10.1097/MLR.0000000000000269.
30. Chaplin B, Meloni S, Eisen G, Jolayemi T, Banigbe B, Adeola J, Wen C, **Reyes Nieva H**, Chang C, Okonkwo P, Kanki P. Scale-up of networked HIV treatment in Nigeria: creation of an integrated electronic medical records system. *Int J Med Inform*. 2015 Jan; 84(1):58-68. PMID:25301692. doi:10.1016/j.ijmedinf.2014.09.006.
31. Linder JA, Doctor JN, Friedberg MW, **Reyes Nieva H**, Birks C, Meeker D, Fox CR. Time of day and the decision to prescribe antibiotics. *JAMA Intern Med*. 2014 Dec; 174(12):2029-31. PMID:25286067; PMCID:PMC4648561. doi:10.1001/jamainternmed.2014.5225.
32. Schiff G, Griswold P, Ellis BR, Puopolo AL, Brede N, **Reyes Nieva H**, Federico F, Leydon N, Ling J, Wachenheim D, Leape LL, Biondolillo M. Doing right by our patients when things go wrong in the ambulatory setting. *Jt Comm J Qual Patient Saf*. 2014 Feb; 40(2):91-6. PMID:24716332. doi:10.1016/s1553-7250(14)40011-4.
33. Linder JA, **Reyes Nieva H**, Blumentals WA. Antiviral and antibiotic prescribing for influenza in primary care. *J Gen Intern Med*. 2009 Apr; 24(4):504-10. PMID:19225847; PMCID:PMC2659164. doi:10.1007/s11606-009-0933-9.

### Forthcoming - Selected in Call for Papers

1. **Reyes Nieva H**, Heck C, Furuya Y, Lee SSJ, Sobieszczyk ME, Zucker J, Castor D. Ethical Frontiers of Agentic AI in Infectious Diseases. *Clinical Infectious Diseases*. Forthcoming.
2. **Reyes Nieva H**, Heck C, Pathela P, Lee SSJ, Sobieszczyk ME, Castor D, Zucker J. Toward a Learning Public Health System: Leveraging Responsible Artificial Intelligence and Informatics for Infectious Disease Surveillance. *Clinical Infectious Diseases*. Forthcoming.

### Manuscripts Submitted to Preprint Servers and/or Under Review

1. **Reyes Nieva H**, Flanagan M, Huang S, Theodore DA, Nkodo AF, Parkinson M, Hill S, McAndrew M, Benitez JA, Peralta H, Amesty S, Zucker JE, Sobieszczyk M, Castor D. Comfort with AI for HIV Prevention Among Cisgender Women in New York City. medRxiv. doi:10.64898/2026.06.02.26354471.
2. **Reyes Nieva H**, Kashyap A, Voss EA, Ostroplets A, Anand A, Ketenci M, DeFalco FJ, Choi YS, Li Y, Allen MN, Guang SA, Natarajan K, Ryan P, Elhadad N. The Impact of Evolving Endometriosis Guidelines on Diagnosis and Observational Health Studies. medRxiv. doi:10.1101/2024.12.13.24319010.
3. Gong L, **Reyes Nieva H**, Yang JY, Aswani N, Wei A, Hur C. Temporal Patterns of Immune-Related Adverse Events Enable Phenotype-Guided Risk and Timing Prediction.
4. Juniu S, Castor D, **Reyes Nieva H**, Charon R, Amesty S. Generative Artificial Intelligence in Medical Education and Participatory Research for Social Action: A Human and AI Comparative Analysis. medRxiv doi: 10.64898/2026.05.14.26351842.
5. Sun TY, Hardin J, **Reyes Nieva H**, Natarajan K, Cheng RJ, Ryan PB, Elhadad N. Large-scale characterization of gender differences in diagnosis prevalence and time to diagnosis. medRxiv. PMID:37873224; PMCID:PMC10592987. doi:10.1101/2023.10.12.23296976.
6. Heck CJ, Quigee D, Ganesan K, Wallach S, Mitchell SL, Farley SM, **Reyes Nieva H**, Howard A, Zucker J, Sobieszczyk M, Wiant S, Justman J, El-Sadr W, Castor D. HIV testing and treatment gaps in subgroups of men with HIV: Pooled findings from seven population-based surveys in eastern and southern Africa.
7. Heck CJ, Quigee D, Wallach S, Zucker J, Ganesan K, Mitchell S, Farley S, Sobieszczyk ME, Howard A, **Reyes Nieva H**, Wiant S, Justman J, El-Sadr W, Castor D. HIV prevalence and recency among subgroups of women with overlapping characteristics in eastern and southern Africa: Findings from seven population-based surveys.

### Scholarship without named authorship

1. Bell SP, Schnipper JL, Goggins K, Bian A, Shintani A, Roumie CL, Dalal AK, Jacobson TA, Rask KJ, Vaccarino V, Gandhi TK, Labonville SA, Johnson D, Neal EB, Kripalani, PILL-CVD Study Group\*. Effect of Pharmacist Counseling Intervention on Health Care Utilization Following Hospital Discharge: A Randomized Control Trial. *J Gen Intern Med*. 2016 May; 31(5):470-7. (\*member of the investigative team cited in the appendix of the manuscript) PMID:26883526; PMCID:PMC4835388. doi:10.1007/s11606-016-3596-3.
2. Kripalani S, Roumie CL, Dalal AK, Cawthon C, Businger A, Eden SK, Shintani A, Sponsler KC, Harris LJ, Theobald C, Huang RL, Scheurer D, Hunt S, Jacobson TA, Rask KJ, Vaccarino V, Gandhi TK, Bates DW, Williams MV, Schnipper JL, PILL-CVD Study Group\*. Effect of a pharmacist intervention on clinically important medication errors after hospital discharge: a randomized trial. *Ann Intern Med*. 2012 Jul 03; 157(1):1-10. (\*member of the investigative team cited in the appendix of the manuscript) PMID:22751755; PMCID:PMC3575734. doi:10.7326/0003-4819-157-1-201207030-00003.

3. Schnipper JL, Roumie CL, Cawthon C, Businger A, Dalal AK, Mugalla I, Eden S, Jacobson TA, Rask KJ, Vaccarino V, Gandhi TK, Bates DW, Johnson DC, Labonville S, Gregory D, Kripalani S, PILL-CVD Study Group\*. Rationale and design of the Pharmacist Intervention for Low Literacy in Cardiovascular Disease (PILL-CVD) study. *Circ Cardiovasc Qual Outcomes*. 2010 Mar; 3(2):212-9. (\*member of the investigative team cited in the appendix of the manuscript) PMID:20233982; PMCID:PMC3021350. doi:10.1161/CIRCOUTCOMES.109.921833.

### *Non-peer reviewed scholarship in print or other media*

1. Huang Y, Aali A, Chen L, Huang X, Jeong C, Jeong H, Kavishwar A, Kumar A, Kondylis JG, Matton K, Nghiem H, Noori A, Paruchuri A, Pradeepkumar J, Sadhuka S, Srivathsa N, Sun T, Unell A, Vaidya A, Celi LA, Fernandopulle R, Joshi S, Mukherjee S, **Reyes Nieva H**, Shanmugam D, Shen L, Singh K, Tang S, Liu Y, Duong Q, Kirmizi C, Chang T, Healey E. Recent Advances, Applications and Open Challenges from ML4H Roundtables]Recent Advances, Applications and Open Challenges in Machine Learning for Health: Reflections from Research Roundtables at ML4H 2025 Symposium. *Proceedings of Machine Learning Research*. 2026.
2. Hegselmann S, Parziale A, Shanmugam D, Tang S, Severson K, Asiedu MN, Chang S, Dossou BFP, Huang Q, Kamran F, Zhang H, Nagaraj S, Oala L, Xu S, Okolo CT, Zhou H, Dafflon J, Ellington C, Jabbour S, Jeong H, **Reyes Nieva H**, Yang Y, Zamzmi G, Mhasawade V, Truong V, Chandak P, Lee M, Argaw P, Heuton K, Singh H, Hartvigsen T. Machine Learning for Health (ML4H) 2023. *Proceedings of Machine Learning Research*. 2023. 225: 1-12. <https://proceedings.mlr.press/v225/hegselmann23a.html>.
3. **Reyes Nieva H**, Kaplan RL, Madani N. Women and HIV: Update from the Middle East and North Africa. Global Network of Researchers on HIV/AIDS in the Middle East and North Africa (GNR-MENA). 2010 International AIDS Conference, Vienna, Austria.

### *Professional educational materials or reports, in print or other media*

1. Interrogating the Ethics of Artificial Intelligence in Healthcare: A Casebook Engaging Justice, Equity, Diversity, and Inclusion\*  
\*member of the writing committee
2. When Things Go Wrong in the Ambulatory Setting. 2013. The PROMISES Study.\*  
\*member of the writing committee
3. Using the Monthly Report Form Tool. 2011. Harvard PEPFAR Nigeria. Training Manual.
4. Using the Adult Quality Improvement Tool. 2011. Harvard PEPFAR Nigeria. Training Manual.
5. Using the Pediatric Quality Improvement Tool. 2011. Harvard PEPFAR Nigeria. Training Manual.
6. Using the Data Quality Assessment Tool. 2011. Harvard PEPFAR Nigeria. Training Manual.
7. Data Quality Control and Assurance (Intermediate Data Manual). 2010. Harvard PEPFAR Nigeria. Training Manual.
8. Guidelines for Reporting Clinical Data Quality Assessments. 2010. Harvard PEPFAR Nigeria. Technical Guideline.
9. Generating Adult Quality of Clinical Care Indicators in an HIV Program. 2010. Harvard PEPFAR Nigeria. Technical Guideline.
10. Generating Pediatric Quality of Clinical Care Indicators in an HIV Program. 2010. Harvard PEPFAR Nigeria. Technical Guideline.

11. Introduction to Data Management (Beginner Data Manual). 2009. Harvard PEPFAR Nigeria. Training Manual.

### *Theses*

1. **Reyes Nieva, H.** Leveraging Computational Methods to Advance Health Equity Science Through Evidence Synthesis, Strategic Monitoring, and Precision Public Health. Doctoral Thesis, Columbia University.
2. **Reyes Nieva, H.** Spatiotemporal Modeling of Opioid Prescribing and Persistent Opioid Use in Eastern Massachusetts, 2000-2018. Master's Thesis, Johns Hopkins Bloomberg School of Public Health.
3. **Reyes Nieva, H.** Individual Development Accounts and Other Asset-based Approaches to Poverty Alleviation. Undergraduate Thesis, Yale University.  
**Received a Mellon Forum Undergraduate Research Award.**

### *Abstracts, Poster Presentations and Exhibits Presented at Professional Meetings*

Works resulting in publication or presented more than 5 years ago are not included below unless awarded special recognition.

### **Regional**

1. **Reyes Nieva H,** Kashyap A, Voss EA, Ostropolets A, Anand A, Ketenci M, Guang S, Allen MN, DeFalco FJ, Choi YS, Li Y, Natarajan K, Ryan P, Elhadad N. Expanding Endometriosis Diagnostic Guidelines Captures More Women with Non-Classical Symptoms, Different Care-Seeking Patterns, and Lack of Access to Surgical Care. Endometriosis: A Look at Current Practices and Emerging Science, New York Academy of Sciences. New York, NY; 06/2025.
2. Flanagan M, Foumena Nkodo A, Parkinson M, Zucker J, Purpura L, Hill S, Wiant S, McAndrew M, Li J, Ahmed S, Peralta H, Benitez J, **Reyes Nieva H,** Theodore DA, Sobieszczyk ME, Castor D. Healthcare Technology Use and Attitudes Toward AI-Assisted Diagnosis Among Individuals With and Without Long COVID in New York City. 2nd Annual Quality in Care Symposium, New York Academy of Medicine. New York, NY; 06/2025.

### **National**

1. Tucker E, **Reyes Nieva H,** Castor D, Yin MT, Gordon P, Lopez K, Zucker J. Doxycycline Post-exposure Prophylaxis among HIV PrEP Users in New York City: Assessing Uptake, Disparities, and STI Outcomes using a Regional Health Information Exchange. ID Week. Washington, DC; 10/2026.
2. Richter LR,\* **Reyes Nieva H,\*** DeLozier S, Arif A, Zucker J. HIV Acquisition After PrEP Initiation: Quantifying Opportunities for Long-Acting Injectable Interventions Using Large-Scale Observational Data. American Medical Informatics Association Annual Symposium. Atlanta, GA; 11/2025.  
**Received the AMIA Distinguished Poster Award**  
**\*Denotes co-first authorship**
3. Hill SE, Parkinson M, Nkodo AF, Ahmed S, Li J, McAndrew M, Flanagan M, Wiant S, Peralta H, Benitez J, **Reyes Nieva H,** Purpura L, Zucker J, Theodore DA, Sobieszczyk ME, Castor D. Social Determinants of Decisions about COVID-19, Influenza, and RSV Immunization Among Predominantly Black and Latino Communities and Households in New York City: A Cross-Sectional Analysis. ID Week. Atlanta, GA; 10/2025.

4. **Reyes Nieva H**, Zucker J, Elhadad N. Elucidating Health Inequities and Research Gaps in HIV and Other Sexually Transmitted Infections Using Data Mining and a Large Language Model: A Computational Scoping Review. Poster Presentation. STI Prevention. Atlanta, GA; 09/2024.
5. D'Avanzo P, Bosco J, Carnevale C, Meyers K, Theodore DA, **Reyes Nieva H**, Zucker J. "I Don't See Myself as a Gatekeeper": Doxy-PEP Knowledge, Attitudes and Practices Among Healthcare Providers in New York. Poster Presentation. STI Prevention. Atlanta, GA; 09/2024.
6. **Reyes Nieva H** and Elhadad N. Mining the Health Disparities and Minority Health Bibliome: A Computational Scoping Review. Oral presentation. American Medical Informatics Association Annual Symposium. Washington, DC; 11/2022.  
**Spotlighted and invited for special presentation by AMIA DEI Committee for "demonstrating best practices in promoting diversity, equity, and inclusion through scholarly communications in biomedical informatics"**
7. Acharya N, **Reyes Nieva H**, Natarajan K. Analyzing shifts in healthcare-seeking behavior among All of Us enrollees in the era of COVID-19. Poster Presentation. American Medical Informatics Association Annual Symposium. Washington, DC; 11/2022.
8. Sun T, Hardin J, **Reyes Nieva H**, Natarajan K, Cheng RF, Ryan P, Elhadad N. Large-scale Characterization of Gender Differences in Age at Diagnosis and Time to Diagnosis in Longitudinal Observational Health Data. National Institutes of Health (NIH) Office of Research on Women's Health (ORWH) Workshop on Gender and Health: Impacts of Structural Sexism, Gender Norms, Relational Power Dynamics, and Gender Inequities. (selected oral abstract presented by Elhadad). Virtual Event; 10/2022.
9. Sun TY, Chen J, **Reyes Nieva H**, Bear Don't Walk OJ IV, Altosaar J, Elhadad N. Gender Differences in Time to Diagnosis through Fairness and Time Variant Evaluation of EHR Data (selected oral abstract presented by Sun). American Medical Informatics Association Annual Symposium. San Diego, CA; 11/2021.
10. Pang C, Chen R, **Reyes Nieva H**, Kalluri KS, Sun TY, Jiang X, Rodriguez VA, Ryan PB, Natarajan K. Characterization and Comparison of Embedding Algorithms for Phenotyping across a Network of Observational Databases. (selected oral abstract presented by Pang). American Medical Informatics Association Annual Symposium. Virtual Event; 11/2020.
11. Boskey E, Tabaac A, Wigell R, Wolf K, Lage I, Landrum S, **Reyes Nieva H**, Bearnot B, Streed C. Using patterns of missing EHR data to identify care disparities in gender diverse patients. (selected oral abstract presented by Tabaac). APHA Annual Meeting and Expo. Virtual Event; 10/2020.

### International

1. **Reyes Nieva H**, Kashyap A, Voss EA, Ostroplets A, Anand A, Ketenci M, DeFalco FJ, Natarajan K, Choi YS, Li Y, Allen MN, Guang S, Elhadad N. The Impact of Evolving Diagnostic Guidelines on Clinical Characterization of Endometriosis. Poster Presentation. OHDSI Global Symposium, New Brunswick, NJ; 10/2024.
2. **Reyes Nieva H**, Zucker J, Tucker EG, McLean J, DeLaurentis C, Gunaratne S, Elhadad N. Development and Validation of Machine and Deep Learning Classifiers for Monkeypox. Poster Presentation. Symposium on Artificial Intelligence for Learning Health Systems (SAIL), Río Grande, Puerto Rico; 05/2023.
3. Desai PM, Ostroplets A, Richter LR, **Reyes Nieva H**, Spotnitz M, Rodriguez VA, Sun TY, Natarajan K. Phenotype Development and Evaluation of Heart Failure: A Case Study in using Patient-Level Prediction to Improve Phenotype Validity. Poster presentation. OHDSI Global Symposium. 09/2021.